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ForCFR, A transverse and Innovative GHG Forecaster to Reduce Net Carbon Intensity for Upstream Activities – Application to WAG CO_2 Projects

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Abstract

TotalEnergies 2030 objective is to achieve a net 40% reduction in scope 1 and 2 greenhouse gases (GHG) emissions compared to 2015. For ADNOC, Carbon management, electrification, energy efficiency and methane reduction are at the core of the strategy to achieve a 25% reduction in GHG intensity by 2030 and be operationally net zero by 2045. CO_2 EOR has the potential to significantly reduce, and potentially fully decarbonize ADNOC operations, especially in onshore oil fields. ADNOC onshore shareholders have the ambition to reach an annual CO2 storage volume of 10 million tons per annum (MTPA) through EOR by 2030. CO_2 can be collected from anthropogenic sources or from oil fields associated gas and injected into oil reservoirs not only to enhance the oil recovery, but also to sequestrate the CO_2within the oil reservoirs.

Estimating the EOR related GHG emissions accurately includes calculating GHG emissions related to the CO_2 recycling operations with the additional oil production treatment, and the amount of stored CO_2 after recycling. The present publication introduces how ForCFR, an internally developed and innovative GHG forecaster (SPE-221930-MS) is used to perform EOR GHG emissions calculation, which can then be used to obtain the net carbon intensity of EOR technologies application. ForCFR is integrating energy consumption and GHG emissions evaluation in production forecasting tools (reservoir modeling softwares). The application of EOR GHG incremental emissions has been done on a preliminary WAG CO_2 project case including the calculation of incremental energy consumption, incremental GHG emissions, CO_2 storage, and, consequently, net oil carbon intensity (negative) and stored CO_2 (positive). In this paper, EOR WAG CO_2 results are discussed in detail and a case on enhancing CO_2 storage in the aquifer below is also presented.

This additional storage benefits from the facilities developed by the WAG CO_2 to enhance CO_2 storage at much lower cost than for a standalone CCS (Carbon Capture and Storage) project. The joint operation of WAG CO_2 and CCS at the same location creates many synergies for operation, monitoring and ensuring the integrity of the CO_2 storage. The results obtained prove that EOR technologies can contribute to the reduction of the overall carbon intensity of oil production. For the EOR CO_2, the stored CO_2 is not only compensating for the full scope 1 & 2 emissions linked to the EOR project, but also reducing by 30% the carbon intensity of the incremental oil production itself (scope 1, 2 & 3).

Thanks to the synergies between CO_2 CCS and EOR CO_2, the additional storage is multiplying by more than two the volume of CO_2 stored, which is then compensating around 40% of the incremental oil production Scope 3 GHG emissions. In addition, the CO_2 storage cost is minimized.

Introduction: What is ForCFR?

ForCFR has been developed within the Zero Carbon Emission Assets R&D program at TotalEnergies. It enables easy forecasting of GHG emissions directly using reference production forecasting tools. ForCFR facilitates the screening of greenhouse gas emissions reduction opportunities, allowing for faster scenario evaluations and decisions based on optimal carbon intensity throughout the asset's life.

The tool was described in a previous ADIPEC presentation in 2024 (SPE-221930-MS) and involves the organic integration of material, heat, and power balance modeling of an asset within a reference hydrocarbon production forecasting software. At every simulation timestep, equipment power needs (e.g., compressors or pumps) are evaluated according to hydrocarbon production and injection potential. These power needs are then transmitted as a load to equipment power sources (e.g., gas turbines) for fuel gas rate evaluation. Flare gas rates are assessed based on flaring policies, combining both planned and unplanned contributions. Finally, fuel and flare gas rates are converted into GHG emissions. Constraints such as asset GHG intensity or condition-driven equipment startups and shutdowns have also been implemented.

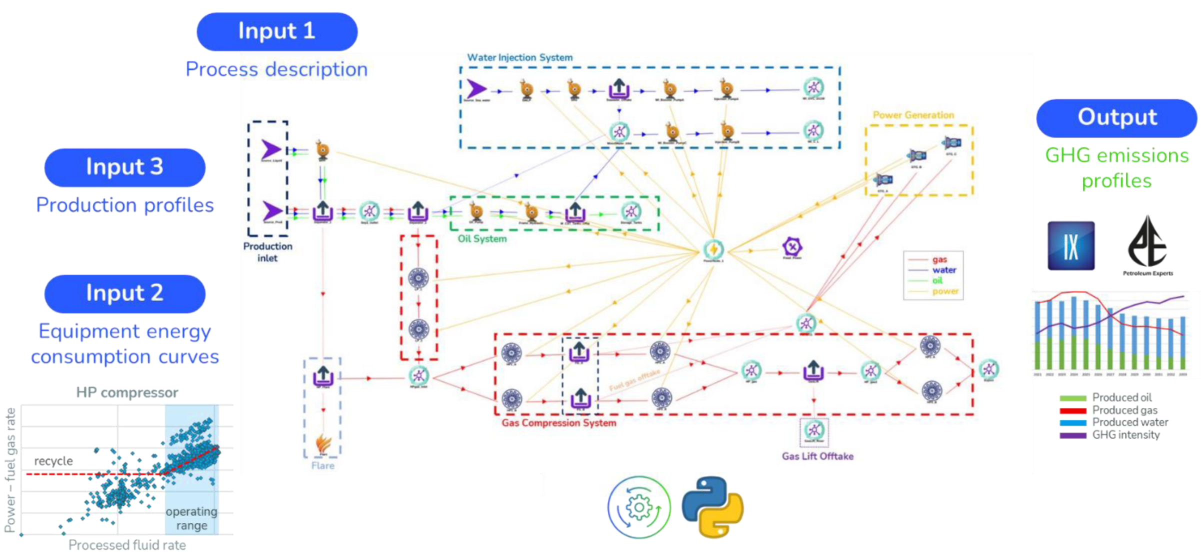

Figure 1—ForCFR workflow

The first step of the workflow is to prepare a simplified process scheme of the oil & gas development plan with the main energy consumers (heaters, coolers, compressors, turbines, main pumps…) and assuming fixed energy consumers for small equipment. This simplified process scheme will be modelized and connected to the reservoir software to issue energy consumption simultaneously with injection and production profiles. Main energy consumers should have their energy consumption curves embedded in the model.

For CFR approach allows screening of various design assumptions from reservoir to surface parameters and promotes sanctioning the most valuable and energy efficient development. Reservoir management can have a significant impact on GHG emissions and shows complex interactions. Long term GHG intensity increases to a higher than initial value, as the reservoir depletes and more energy is required to sustain production (e.g. riser gas-lift) and to treat associated produced gas and water. The tool can quantitatively orientate the technical choice to maintain a high level of production without increasing GHG intensity. It allows to better and faster capture subsurface - surface interactions and to evaluate more GHG intensity reduction opportunities with associated hydrocarbon production, leading to more asset evaluation all along field life. In particular, the tool demonstrated the efficiency of EOR techniques to reduce GHG intensity. For example, polymer flooding increases oil mobility towards water mobility, reducing water production. When water production is reduced, short term GHG intensity decreases as less water is produced, treated and pumped for re-injection. WAG (Water Alternate Gas injection) was also simulated and shows steady increase of GHG emissions with increasing volume of gas to be compressed due to GOR increase, but the WAG efficiency was clearly seen with incremental oil production ultimately reducing the GHG intensity. For EOR cases, base line (usually water flood or secondary recovery using pressure maintenance) should be computed as well to clearly see the incremental oil production vs the incremental GHG emissions relative to the addition of facilities (compressors, desalination units, polymer units…). Any EOR techniques involve the injection of an EOR 'agent' which will microscopically or macroscopically positively impact the oil mobility and recovery in the reservoir. This agent also usually partially reproduces with oil, water and gas production (gas, polymer, surfactant) and asks for additional facilities to treat these products. All these additional energy consumers (injection and production facilities) should be integrated in the simplified process and oil production, water injection and production profiles, GHG emissions and GHG intensity profiles will be compared between base case and EOR case.

Application of ForCFR for GHG emissions forecasting on WAG CO_2 project

ForCFR was applied to a WAG CO_2 project, with or without the addition of additional CO_2 storage thanks to CCS in a related aquifer.

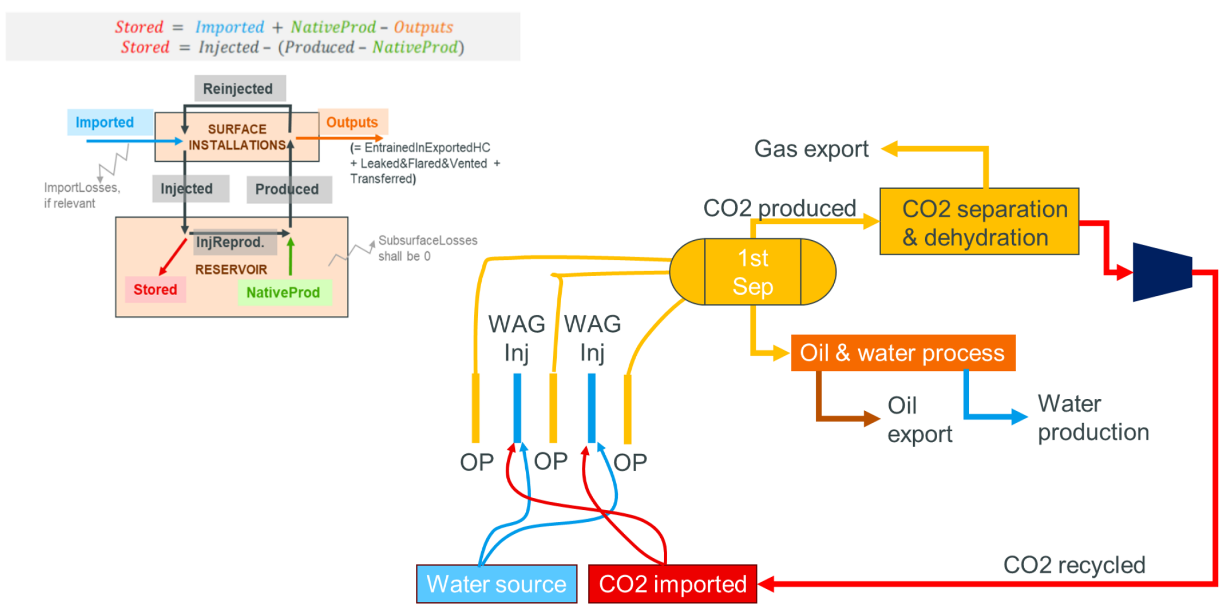

Figure 2—WAG CO_2 project simplified process

The WAG CO_2 simplified process is shown above. Initial injected CO_2 will come either from an imported external source (we call it merchant CO_2) or/and from a native source of CO_2 already present in the reservoir which will be recycled in the reservoir as an EOR agent (we call it CFR). The above material balance shows the different flows of CO_2 in a WAG CO_2 cycle: imported and recycled CO_2 are injected into the reservoir. Some CO_2 is stored in the reservoir or migrated (Subsurface losses), a part of the CO_2 injected will be reproduced along with incremental oil production. The produced gas stream could be exported, and its CO_2 content will be considered as output CO_2. CO2 separated from produced gas will be recycled and reinjected in the reservoir. WAG CO_2 projects usually leave a minimum of CO_2 in the export gas to minimize the CO_2 separation unit cost and meet the export gas specifications.

The stored CO_2 volume has become an important parameter for TotalEnergies and other IOCs and NOCs as the WAG CO_2 is an efficient way to store CO_2 in addition to its well-known efficiency as EOR technique.

- TotalEnergies plan to increase its energy production (oil, gas and electricity) overall, by 4% per year between 2024 and 2030
- Meanwhile, TotalEnergies 2030 objectives are to net reduce by 40% the scope 1+2 (compared to 2015)
- TotalEnergies is focusing its project investments on oil prospects with low technical costs, low GHG emissions (intensity < 16 $kgCO_2/bbl$) and short lead times.
- EOR (Enhanced Oil Recovery) will contribute to reduce GHG emissions intensity from operated facilities

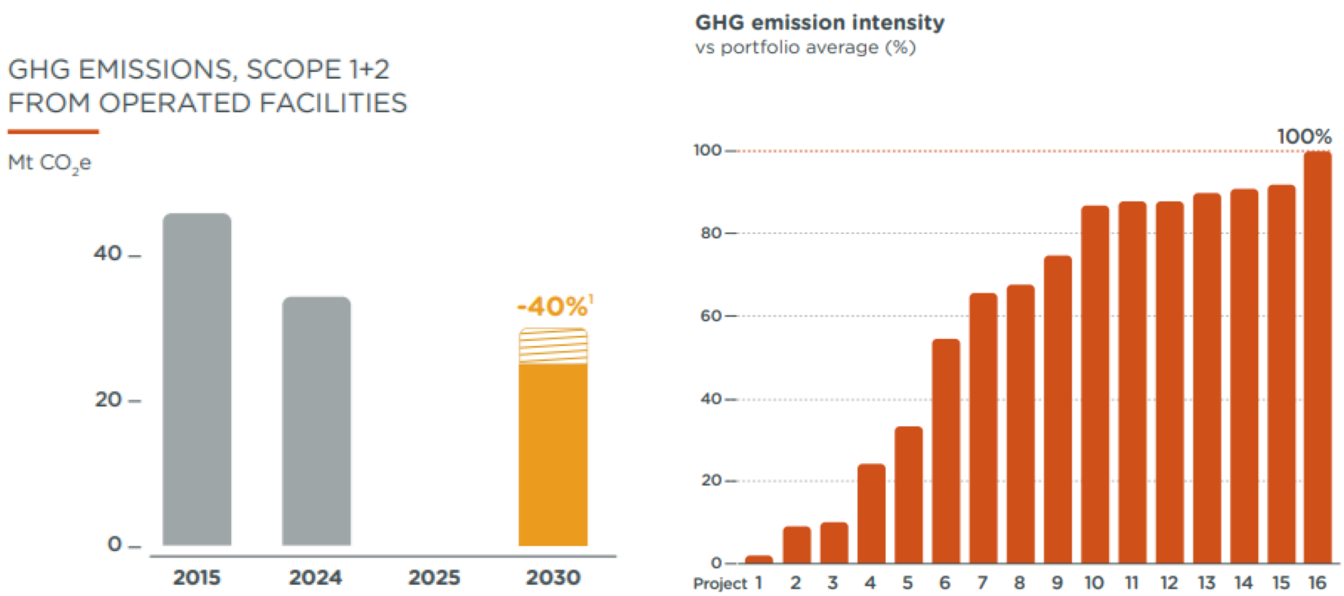

Figure 3—GHG emissions reduction plan (scope 1 & 2) for TotalEnergies up to 2030 (Sustainability & Climate 2025 Progress Report)

WAG CO2 is an important EOR technology to achieve TotalEnergies' GHG intensity reduction target and global Scope 1 and 2 reduction while increasing energy production. As the amount of CO2 stored will be well above the Scope 1 and 2 emissions of the oil and gas production site, we can also look for Scope 3 reduction thanks to EOR CO2. Additionally, a WAG CO2 project is ultimately a CO2 storage project, as the imported CO2 will be stored at the end of the oil and gas production project.

WAG CO2 Coupled with Additional CO2 Storage in Aquifers

As mentioned above, EOR CO2 is an efficient way to store CO2 but has a limitation. The capacity to store CO2 in an oil reservoir decreases over time due to the depletion of the oil bank and the increase in CO2 production alongside oil. WAG CO2 projects are generally operated with a fixed CO2 injection rate into dedicated WAG injectors. Initially, this CO2 comes entirely from an external (imported) source. In the early stages, the injected CO2 is fully retained within the reservoir, and CO2 production is negligible. However, after several years, depending on reservoir heterogeneity and well spacing, CO2 begins to be produced along with oil.

This reproduced CO2 is then recycled and reinjected into the reservoir, which gradually reduces the need for imported CO2. As the oil bank depletes, CO2 production increases. A second stabilization phase in CO2 storage typically occurs when injection is extended into new reservoir zones. In CO2-EOR projects, the main objective is oil production and economic return. This drives the optimization of CO2 utilization, aiming to maximize oil recovery while minimizing total CO2 use. Over time, recycled CO2 becomes sufficient to meet injection needs, and the import of CO2 can be stopped

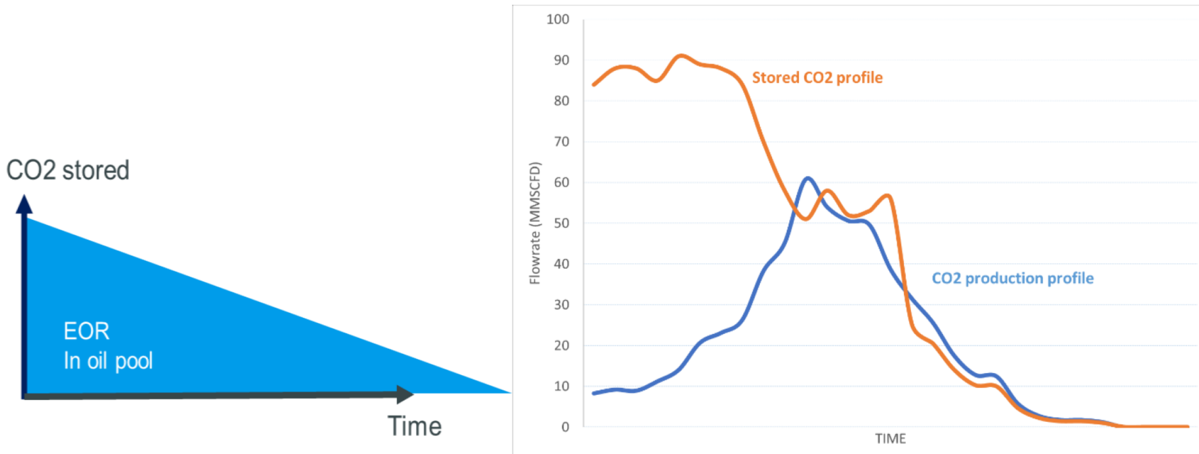

Figure 4—Evolution of stored CO2 daily volume with time

Consequently, the stored CO2 volume of a given WAG CO2 project will be limited to 60-80% of the total injected CO2. As more CO2 is reproduced and recycled over time, the need for imported CO2 gradually declines. From an industrial emitter's point of view, this decline in CO2 storage volume is not favorable. Additionally, during any shutdown of the EOR operation, CO2 injection and therefore CO2 storage are stopped. From the EOR CO2 operator's perspective, the facilities installed to transport, receive, meter, and compress (pump) the imported CO2 for injection are designed for the maximum initial CO2 injection rate (90 MMSCFD in the above example), which is used only for a few years. The EOR CO2 operational uptime will also be impacted by CO2 export shutdowns from its CO2 supplier (the industrial emitter). Developing a CO2-EOR project jointly with a CCS system brings several valuable synergies, as shown in the figure below.

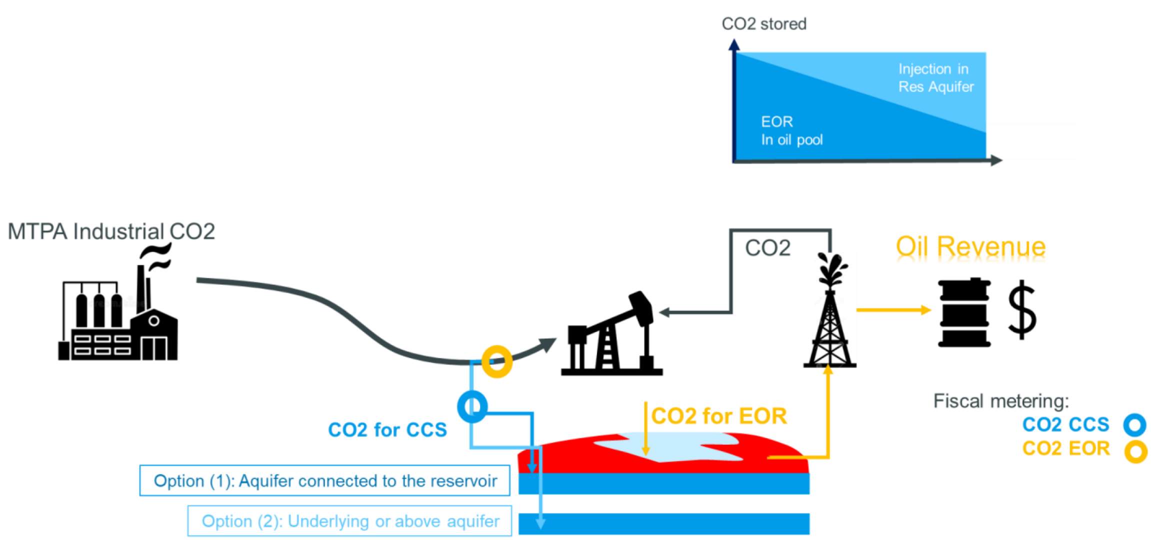

Figure 5—WAG CO_2 project coupled with CO_2 storage in aquifer

The CO2 transportation pipeline from the industrial emitter will still be designed according to the initial CO2 injection flow rate in the oil pool, but this capacity will be used permanently to feed both EOR and

CCS operations. As EOR CO2 demand declines, CO2 injection for storage will increase, and ultimately, EOR CO2 could be stopped while CCS operations continue, transforming the oil and gas site into a CO2 storage site. Additional CO2 emitters could also be connected to the CO2 pipeline, providing flexibility and a steady CO2 supply for both EOR and CCS. EOR and CCS operations will be fully independent, except for the importation of industrial CO2: the shutdown of EOR will not necessarily impact CCS and vice versa. With this scenario, industrial emitters have a steady and constant CO2 storage rate while EOR CO2 has more CO2 availability.

To be considered a CCS project and possibly receive financial support through carbon tax credits or subsidies, CO2 sent to storage in the aquifer must be metered and injected from injection wells dedicated to CCS, and the aquifer should be physically separated from the oil pool. An underlying or overlying aquifer will meet these requirements. Using an aquifer connected to the reservoir can bring operational advantages such as pressure maintenance, which can produce incremental oil, but there is the possibility that this CCS CO2 storage will be considered as IOR/EOR. In any case, it is recommended to meter separately the EOR CO2 being injected into the oil pool using WAG wells and the CCS CO2 going to the aquifer for storage.

Results from GHG Emissions Calculation for a WAG CO2 Project with or without Coupled CCS

For a given WAG CO2 project in the Middle East, where the aquifer below is in contact with the oil pool, the base case is water-flood without gas injection. The WAG CO2 injection starts in 2028, and simulations are run up to 2100. The ForCFR methodology was applied to calculate GHG emissions profiles. Oil production increases with the implementation of WAG CO2 by 81 MMbbls (in 2100), while moderate pressure support offered by the CO2 storage in the aquifer brings an additional 3 MMbbls of incremental oil. Water injection in the base case is replaced by CO2 injection in the WAG case, reducing water injection. Additionally, some water is produced from the aquifer to maintain pressure during CCS, which further reduces water injection in the WAG CO2 + CCS case.

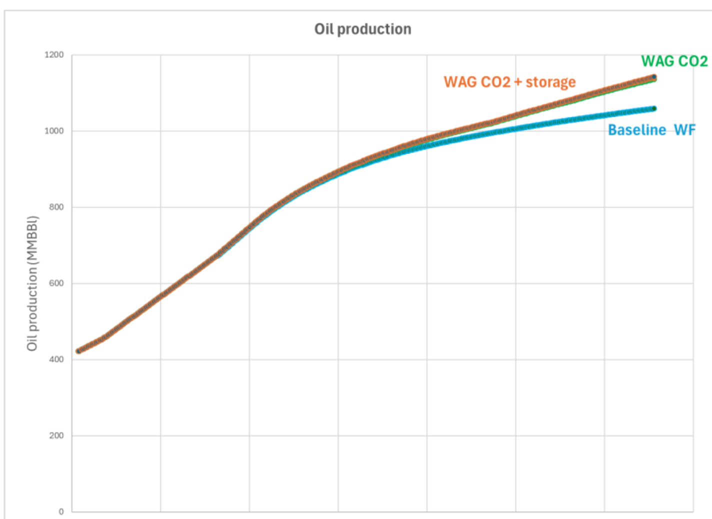

Figure 6—Oil production profiles for the WAG CO_2 case with or without CO_2 storage in aquifer

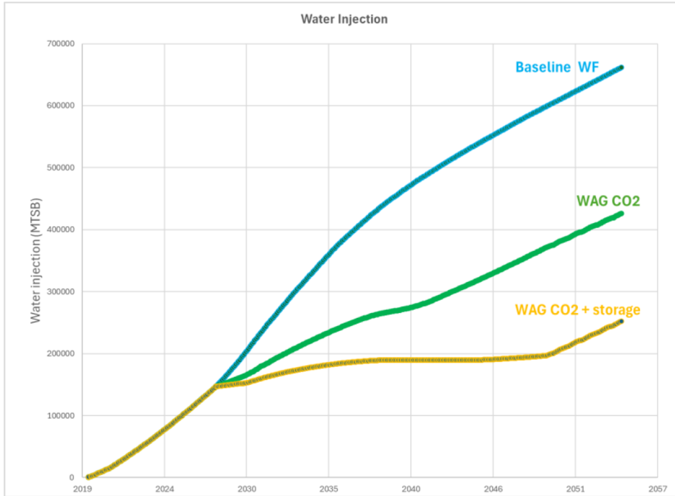

Figure 7—Water injection for WAG CO_2 case with or without CO_2 storage in aquifer

Let's first look into Scope 1 and Scope 2 emissions of the project without considering the volume of stored CO2. Due to the implementation of additional processes to produce significant gas volumes, separate, and recycle the CO2 in the WAG CO2 case, global energy consumption and GHG emissions will increase compared to the base case. When calculating the GHG intensity (GHG emissions divided by hydrocarbon production), we can see from the figure below that the GHG intensity of the WAG CO2 case is lower than that of the base case, which demonstrates an interesting effect of EOR: the reduction of site emissions intensity. In the long term (by 2048, 20 years after WAG CO2 implementation), the decrease in WAG efficiency (less oil production per given volume of CO2 injection) will cause the GHG intensity of the WAG CO2 case to surpass that of the base case water-flood. Economically, the WAG CO2 operation will be stopped (CO2 injection halted) when the cost of CO2 separation and recycling is no longer compensated by the incremental oil recovery.

The GHG intensity of the WAG CO2 + storage case is higher than that of the WAG CO2 only case. The additional energy consumption required to inject more CO2 is not offset by sufficient incremental oil production, which was expected. CCS does not bring any energy savings to the project but does result in more CO2 storage.

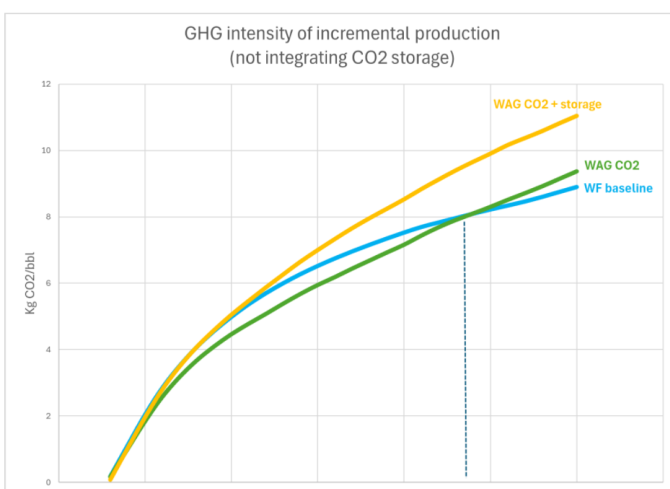

Figure 8—GHG intensity of incremental oil production (not integrating CO_2 stored)

Let's now integrate the stored CO2 into the CO2 balance to highlight the benefit of CCS. In the following graph, the net GHG emissions are presented: atmospheric Scope 1 and 2 GHG emissions (as kgCO2eq) minus the CO2 stored in the reservoir and aquifer. The water flood baseline Scope 1 and 2 GHG emissions are a few million tons of CO2. CO2 storage thanks to WAG CO2 fully compensates for the Scope 1 and 2 emissions, with the stored volume reaching 40 MMT by 2100. The addition of CO2 storage in the aquifer will more than double the CO2 storage, reaching 88 MMT. The global balance of the WAG CO2 + storage case is an incremental oil production of 84 MMbbls and 88 MMT of CO2 stored. The average volume of CO2 stored for the EOR case is 0.5 MTPA, while WAG CO2 + storage is 1.2 MTPA (which is more than the Quest CCS project in Canada).

Figure 9—GHG emissions from water flood base case and WAG CO_2 with or without storage in aquifer

One way to valorize the net CO2 storage achieved by the WAG CO2 + storage project is to compensate for the future emissions resulting from the use of the incremental oil production for car transportation (Life Cycle Analysis, LCA – from well to wheel). A global GHG emissions hypothesis of 500 kgCO2/bbl was used to account for the emissions integrating the carbon content of the oil barrel during transportation and refining of oil on one side, and the distribution and combustion of gasoline on the other side. The comparison of the CO2 stored to the CO2 emitted by the LCA of the incremental produced oil barrel is shown below.

As shown before, the CO2 emissions (Scope 1 and 2) from the site facilities are minimal compared to the volume of stored CO2. WAG CO2, combined with additional storage, significantly increases the amount of CO2 stored and compensates more for the incremental oil production Scope 3 emissions: WAG CO2 will compensate for 30%, while WAG CO2 with additional storage in the aquifer will compensate for around 45% of the Scope 3 emissions from the incremental oil production.

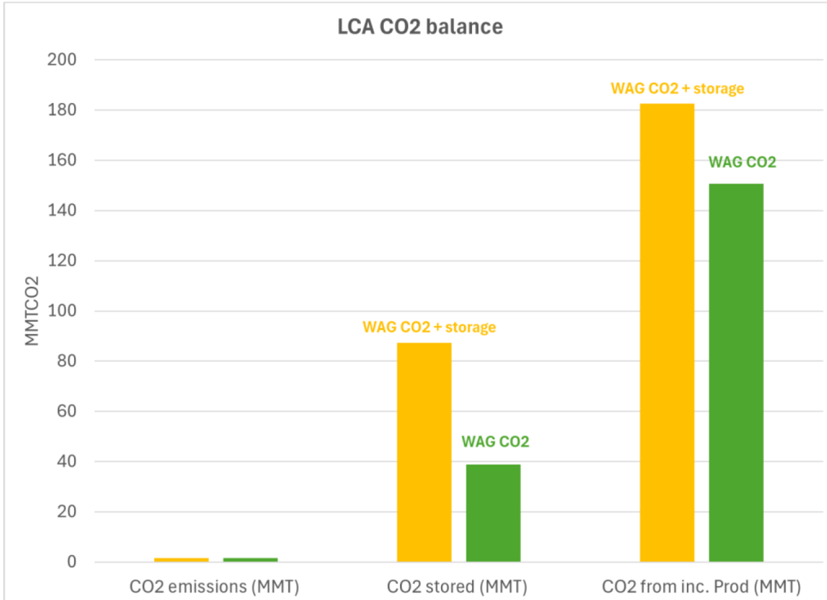

Figure 10—CO_2 emissions (scope 1 &2) compared to CO_2 stored and CO_2 issued (scope 3) from incremental oil production

Conclusions

IOCs and NOCs have set objectives to drastically reduce the GHG intensity of their oil and gas production soon. TotalEnergies' 2030 objectives are to achieve a net reduction of 40% in Scope 1 and 2 GHG emissions (compared to 2015) and to have a GHG intensity below 16 kgCO2/bbl. ForCFR software has been developed at TotalEnergies and enables easy forecasting of GHG emissions directly within reference production forecasting tools. ForCFR was used on a WAG CO2 project with or without additional CO2 storage. Without integrating the CO2 storage in the calculation, it has been proven that EOR reduces the GHG intensity of incremental oil production to less than 15 kgCO2/bbl, thanks to higher oil production and lower water production compared to the base case. Among EOR techniques, WAG CO2 is a very powerful method as it brings significant incremental oil production and stores industrial CO2, resulting in negative site emissions. There are significant operational synergies in performing EOR CO2 and CCS on the same site. EOR CO2, such as WAG CO2, will be performed by injection into the oil pool, while CCS will inject into an aquifer. WAG CO2 combined with additional storage in an aquifer more than doubles the amount of CO2 stored and can compensate for up to 40% of Scope 3 GHG emissions from the incremental oil production.

Nomenclature

CCS: Carbon Capture and Storage
CCUS: Carbon Capture, Utilization and Storage
EOR: Enhanced Oil Recovery
IOC: International Oil Companies
NOC: National Oil Companies
GHG: Green House Gas
GHG intensity: Green House gas emissions (kTCO2eq) divided by oil production (Kbbls) for the same time duration
WAG: Water Alternate Gas
IMPORTED CO_2: CO_2 from an external source (industrial emissions)
OUTPUT CO_2: CO_2 which is exported, entrained, transferred in the produced gas or fuel gas
NATIVEPROD: CO_2 initially present in the reservoir before WAG CO2 implementation

STORED CO_2: CO_2 ultimately stored in the reservoir, which is usually close to IMPORTED CO2 volume

Reference

SPE-221930 For CFR - Account for Greenhouse Gases Emissions in Upstream ActivitiesThrough Sustainable Integrated Asset Modeling, R. Marmier, P. Chehade, M. Akchiche, J. Pellegrino, M. Shmeis, A. Matusse, C. Dauba, and M. Higelin, TotalEnergies, France, ADIPEC 2024, held in Abu Dhabi, UAE, 4 – 7 November 2024.